

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0331

V C Summer 138 kV
May 22, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0331. The Interconnection Customer proposes a 61.3 MW solar photovoltaic facility interconnecting at the V C Summer 138 kV substation in the Pennsylvania area. The request was received on May 05, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 153 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,610,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Parleys-Manati 8401 138 KV
Contingency: Medoa-P.T.S 345 KV line
Loading: 159.7% (Rating: 204 MVA)
DFAX: 10.4%
Pre-project loading: 156.6%
2. Facility: Parleys-Gregg Substation 138 KV
Contingency: P.T.S-Medora 345 KV L
Loading: 131.5% (Rating: 229 MVA)
DFAX: 10.4%
Pre-project loading: 128.7%
3. Facility: Gregg-Dravosburg 138 KV L
Contingency: Medora-P.T.S 345 KV
Loading: 123.6% (Rating: 237 MVA)
DFAX: 10.4%
Pre-project loading: 121.0%
4. Facility: Sunrise Ctr-Lutesville 345 KV
Contingency: Cayucos-Sunrise Ctr 500 KV
Loading: 116.8% (Rating: 1,202 MVA)
DFAX: 5.7%
Pre-project loading: 116.6%
5. Facility: Edes Tap 2-West Campus 345 KV Line
Contingency: Orchard Road-Shawville Station 500 KV
Loading: 115.4% (Rating: 1,155 MVA)
DFAX: 5.6%
Pre-project loading: 115.1%

6. Facility: Edes Tap 2-Lutesville 345 KV Line
 Contingency: Orchard D-Shawville Sta Substation 500 KV line
 Loading: 111.6% (Rating: 1,191 MVA)
 DFAX: 5.4%
 Pre-project loading: 111.4%
7. Facility: Wilmerding-Dravosburg 138 KV L
 Contingency: Medora-P.T.S 345 KV
 Loading: 109.3% (Rating: 252 MVA)
 DFAX: 10.4%
 Pre-project loading: 106.8%
8. Facility: Wahiawa-North Allegheny Windpower Project Substation 345 KV line
 Contingency: Sunrise Ctr-Merret 500 KV
 Loading: 109.1% (Rating: 1,447 MVA)
 DFAX: 6.2%
 Pre-project loading: 108.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Parleys-Manati 8401 138 KV	Rebuild line with higher-capacity conductor	\$1,515,000	100% (\$1,515,000)
Parleys-Gregg Substation 138 KV	Rebuild line with higher-capacity conductor	\$511,000	100% (\$511,000)
Gregg-Dravosburg 138 KV Line	Rebuild line with higher-capacity conductor	\$3,231,000	100% (\$3,231,000)
Sunrise Ctr-Lutesville 345 KV	Rebuild line with higher-capacity conductor	\$37,850,000	2% (\$607,000)
Edes Tap 2-W Campus 345 KV	Rebuild line with higher-capacity conductor	\$22,395,000	2% (\$357,000)
Edes Tap 2-Lutesville 345 KV Line	Rebuild line with higher-capacity conductor	\$54,361,000	2% (\$884,000)
Wilmerding-Dravosburg 138 KV L	Reconductor line	\$2,342,000	100% (\$2,342,000)
Wahiawa-North Allegheny Windpower Project Substation 345 KV line	Reconductor line	\$9,218,000	2% (\$163,000)
	Total allocated to project		\$9,610,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.